

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program,
then use `chmod`.

Create a file called `permission.c`

```
#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

Compile

```
gcc permission.c -o permission
```

Run

```
./permission
```

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

chmod 666 sample.txt **Check** `ls -l sample.txt` You should see
`-rw-rw-rw-` 1 user user ... sample.txt

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt
```

```
chmod o+rw sample.txt
```

```
chmod a+rw sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing
→ 600

9) Owner: read + write; Group: read; Others: read

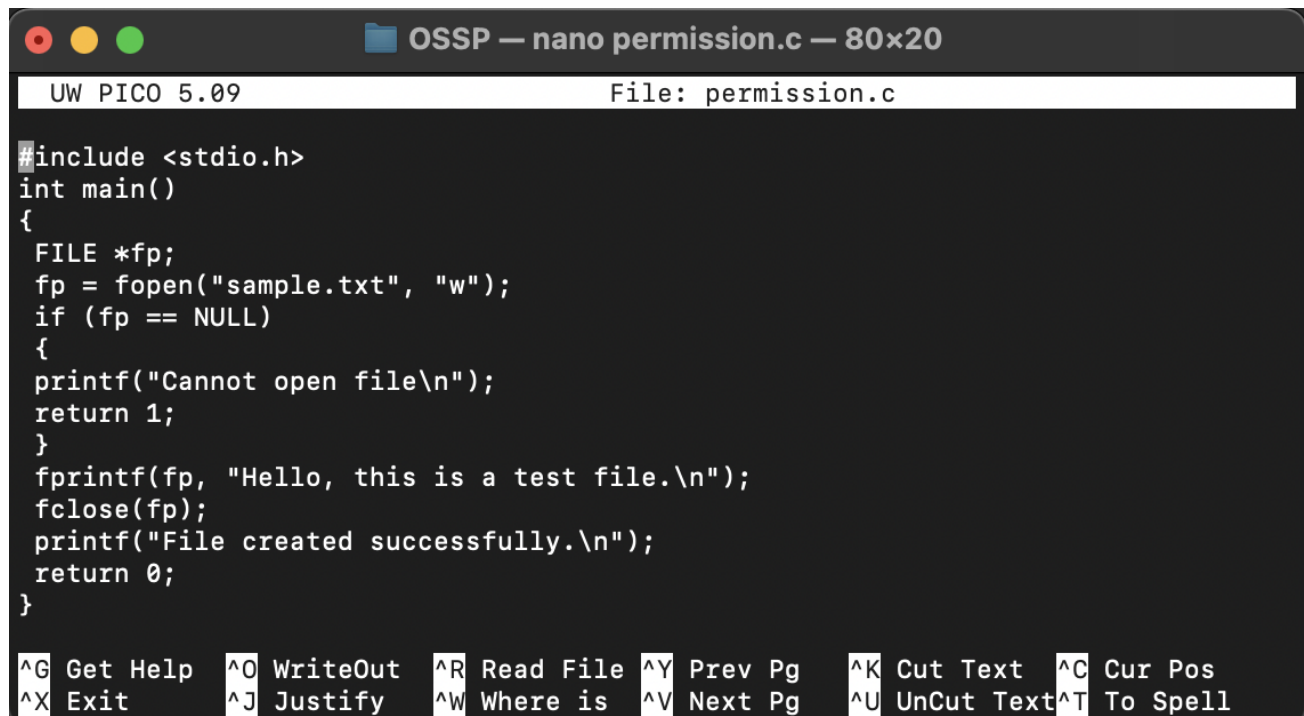
→ 644

10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755

11) Owner: read + write; Group: read + write; Others:
nothing → 660

Everyone: read only → 000

OUTPUT



```
UW PICO 5.09                               File: permission.c

#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

^G Get Help	^O WriteOut	^R Read File	^Y Prev Pg	^K Cut Text	^C Cur Pos
^X Exit	^J Justify	^W Where is	^V Next Pg	^U UnCut Text	^T To Spell

```

[apple@apples-MacBook-Pro-2 OSSP % chmod 600 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw----- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 644 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw-r--r-- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 755 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rwxr-xr-x 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 600 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw----- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 660 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw-rw---- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 000 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
----- 1 apple  staff  28 Aug 17 10:48 sample.txt
apple@apples-MacBook-Pro-2 OSSP % Z

```

TASK OF THE DAY OUTPUTS:-

```

[apple@apples-MacBook-Pro-2 OSSP % chmod 600 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw----- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 660 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw-rw---- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 600 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-rw----- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 444 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
ls: sample.txtt: No such file or directory
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-r--r--r-- 1 apple  staff  28 Aug 17 10:48 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % chmod 440 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
-r--r----- 1 apple  staff  28 Aug 17 10:48 sample.txt
apple@apples-MacBook-Pro-2 OSSP % 
[apple@apples-MacBook-Pro-2 OSSP % chmod 100 sample.txt
[apple@apples-MacBook-Pro-2 OSSP % ls -l sample.txt
---x----- 1 apple  staff  28 Aug 17 10:48 sample.txt
apple@apples-MacBook-Pro-2 OSSP % 

```